

INNOVA JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

--

CG

--	--	--	--	--

INDEX NUMBER

--	--	--	--

BIOLOGY

9744/04

Paper 4 Practical

16 August 2018

2 hours 30 minutes

Candidates answer on the Question Paper.

READ THESE INSTRUCTIONS FIRST

Write your name, CG and index number on page 1, 7 and 11.
Give details of the practical shift and laboratory, where appropriate, in the boxes provided.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams, graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate. You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

Shift
Laboratory

For Examiner's Use	
1	25
2	19
3	11
Total	55

This document consists of **16** printed pages.



Answer **all** questions.

Question 1

A business has been buying milk from the same supplier for a number of months. Recently, the business has found that the milk has been diluted with water. Milk contains macromolecules like proteins which are denser than water thus milk sinks when placed in aqueous solutions.

- (a) Predict the behaviour of a milk droplet when placed in water with respect to milk's water content.

[1]

The amount of water added to a milk sample can be determined by measuring the density of the milk using aqueous solutions like copper sulfate solution of a standard concentration. When a small drop of milk is placed in copper sulfate, a layer of copper proteinate forms around the milk and this prevents the milk and copper sulfate solution mixing.

Fig. 1.1 shows the movement of a drop of milk through the copper sulfate solution.

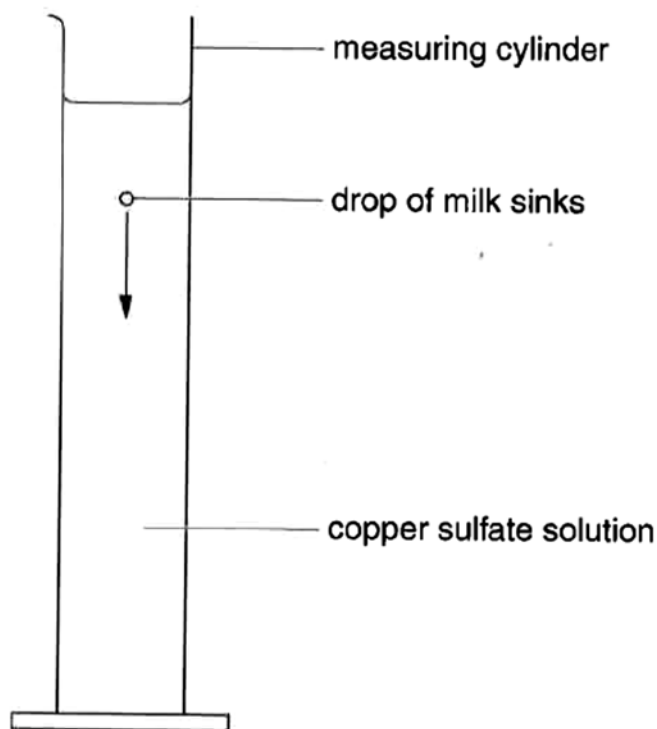


Fig. 1.1

You are required to estimate the percentage of water added to the milk supplied to the business.

You are provided with

- 100% milk, labelled **M**
- milk sample supplied to business, labelled **B**
- distilled water, labelled **W**
- 0.03 mol dm^{-3} copper sulfate, labelled **C**

You are advised to read through the entire procedure before beginning the experiment.

- 1 Prepare 10 cm³ each of a suitable number of concentrations of milk to help you in your investigation. Record the volume of 100% milk, **M** and distilled water, **W** used in your preparation in a table below.

[3]

- 2 Using the syringe with attached needle, release one drop of **M** into **C** in a measuring cylinder. Record the time taken by the droplet to sink in an appropriate format in the space provided below.

Note: Needle attached to syringe is sharp. Handle with care. Keep needle capped when not in use.

Observe the largest fragment of **M** should the droplet break up in the copper sulfate solution.

[3]

- 3 Repeat **step 2** for all milk concentrations you have prepared in **step 1**. You may reuse the copper sulfate unless the milk residue obstructs your vision.
- 4 Repeat the procedure to obtain a **total of 2 replicates**. Perform appropriate calculations on your readings. [1]
- 5 Describe how you would carry out **step 2** to increase the accuracy of your observations.

.....

.....

.....

.....

.....

..... [3]

- 6 Calculate the speed at which the milk droplets sank in distance per unit time. Include your calculations in **step 2**. [2]
- 7 Estimate the percentage of water added to the milk sample supplied to business, **B**. Explain how you derived at your answer.

percentage of water added [1]

Explanation

.....

.....

..... [2]

- 8 Describe **one** way to improve your estimate in terms of

(a) reliability;

.....

..... [1]

(b) accuracy.

.....

..... [1]

Further investigation was conducted to find the protein concentration in sample **B** using biuret's test. The absorbance by sample **B** was measured using a colorimeter and compared to a range of protein solutions of known concentrations. Table 1 shows the absorbance by the protein solutions.

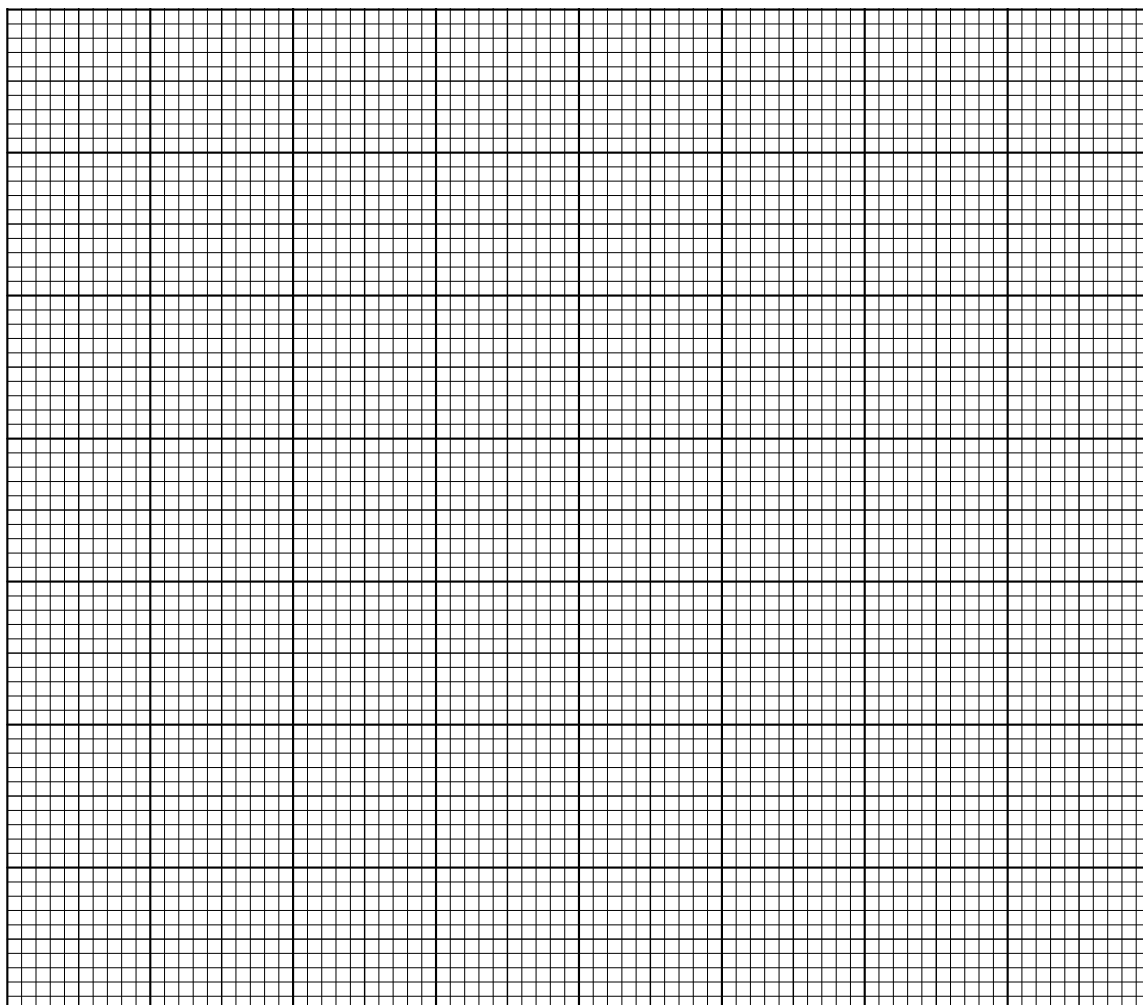
Table 1

Protein concentration / %	Absorbance / arbitrary units
100	65
80	55
60	38
40	21
20	10

- 9 Describe how you would conduct biuret's test using 1 cm³ of milk sample.

----- [2]

- 10 Plot a suitable graph using data provided in Table 1.



- 11** The absorbance of the milk sample **B** was recorded to be 26 arbitrary units. Using your graph, deduce the protein concentration in the milk sample **B**. Show on your graph, how you arrived at your answer.

protein concentration of milk sample, **B** [2]

[Total: 25]

Question 2

You are provided with a microscope slide of a cross section of monocotyledonous stem, labelled **K1**. Fig. 2.1 shows an outline of the stem.

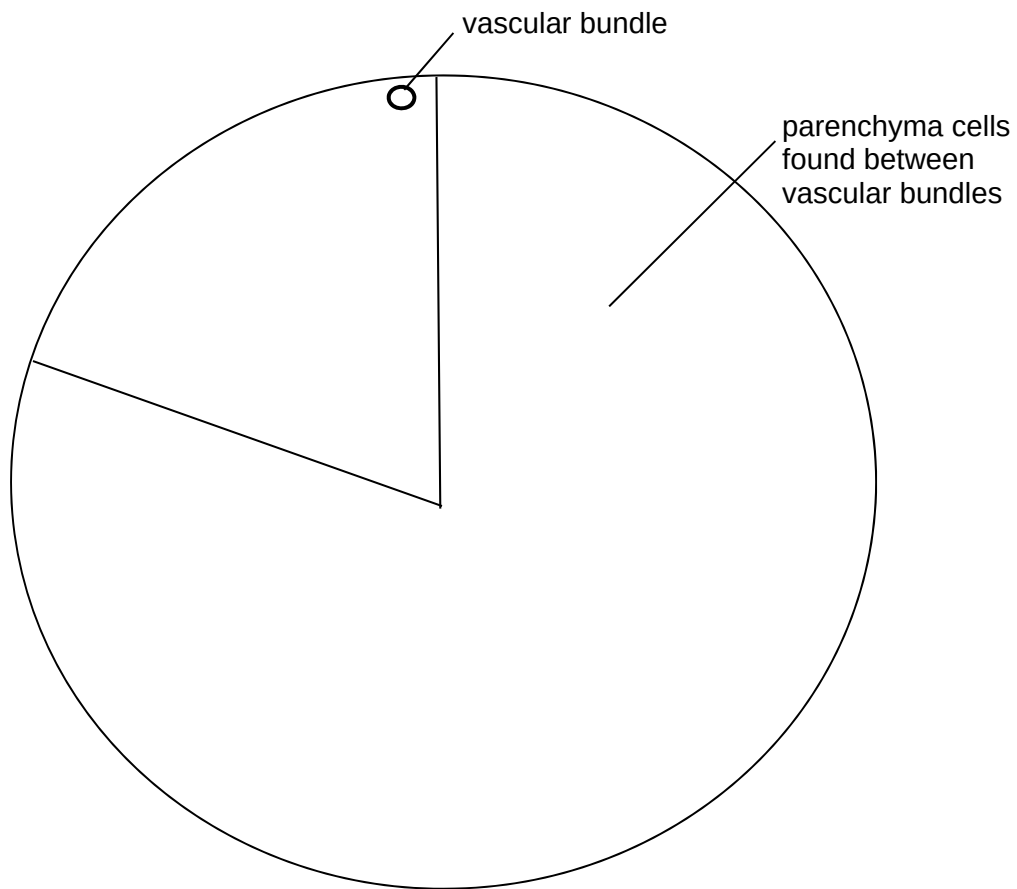


Fig. 2.1

- 1** View **K1** using the **X4 objective** lens. Indicate the distribution of vascular bundles in one fifth of the stem in Fig. 2.1 using open circles, **O**. One vascular bundle is shown in Fig. 2.1. [2]
- 2** You are provided with a stage micrometer. Each division measures 0.01mm. Using the stage micrometer, estimate the diameter of the field of vision under **X40 objective** in **mm**.

diameter of field of vision [1]

- 3** Using your answer in **step 2**, calculate the diameter of the field of vision under **X10 objective** in **mm**. Show your working.

diameter of field of vision [1]

- 4 Count the number of vascular bundle in the field of vision viewed using **X10 objective** lens. Avoid the centre of the specimen.

number of vascular bundles [1]

- 5 Calculate the density of vascular bundles in **K1** using the formula

$$\text{Density of vascular bundle} = \frac{\text{Number of vascular bundle}}{\text{Area of field of vision}}$$

Show any other calculations you performed to obtain the answer.

density of vascular bundles [1]

- 6 Comment on the reliability of your answer in **step 5**.

.....

 [2]

- 7 View a typical vascular bundle using high power objective. State the objective lens you used.

objective lens [1]

8 Draw, **to the same scale**, **one** typical cell from each of the following type

(a) phloem;

Magnification = (show your working below)

Answer = [3]

(b) parenchyma.

Magnification = (show your working below)

Answer = [3]

9 (a) Indicate the **actual size** of the cells drawn in **8(a)** and **(b)**, in μm . [2]

(b) Calculate the magnification of your drawings and show your working clearly in **8(a)** and **(b)**. [2]

[Total: 19]

Question 3 starts on Page 11

Question 3

Many people are intolerant to the disaccharide lactose, which is found in milk. The enzyme lactase is used commercially to catalyse the breakdown of lactose to the monosaccharides glucose and galactose. These sugars taste sweeter and are easier to digest than lactose.

Enzymes can be immobilised in a number of different ways, using different materials.

Fig. 3.1 shows three ways of immobilisation of enzymes.

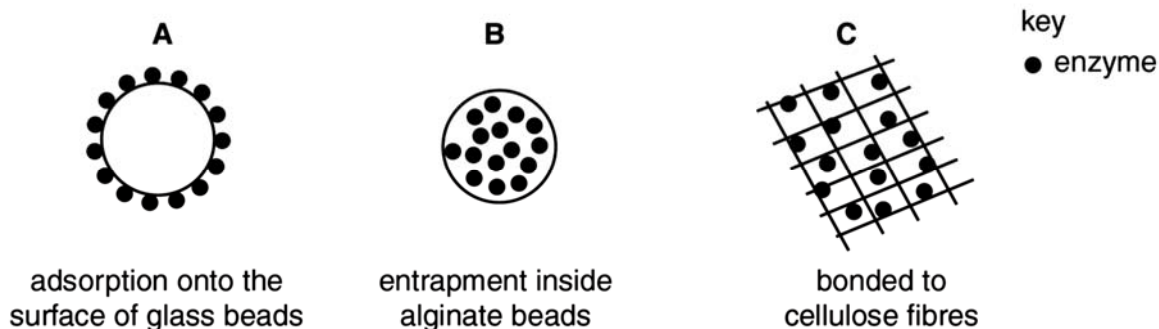


Fig. 3.1

A student carried out an investigation to compare the activity of the enzyme lactase that had been immobilised in the three different ways shown in Fig. 3.1. The enzyme can be reacted with lactose by pouring lactose solution through a column containing the immobilised enzyme. The subsequent glucose produced can be measured using a biosensor (Fig. 3.2) that can measure the concentration of glucose in a solution in $\text{mg}/100\text{cm}^3$.

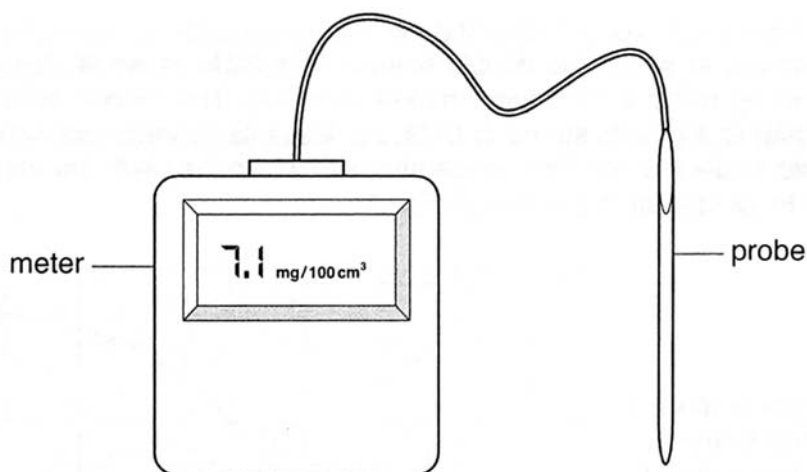


Fig. 3.2

Outline a procedure the student could use to compare the activity of lactase that has been immobilised in different ways.

You are provided with and must use:

- lactose solution
- lactase immobilised using glass beads, **A**
- lactase immobilised using alginate beads, **B**
- lactase immobilised using cellulose fibres, **C**
- biosensor that measures glucose concentration
- stopwatch

You may select from the following apparatus and use appropriate additional apparatus:

- normal laboratory glassware e.g. test-tubes, beakers, measuring cylinders, graduated pipettes, glass rods, etc.
- syringes

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- be illustrated by relevant diagrams, if necessary,
- identify the independent and dependent variables,
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- show how you will record your results and the proposed layout of results tables and graphs,
- apply appropriate statistical test to support conclusion,
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[Total: 11]

